

Cover story **CANCER**

radiation, he asked the doctors. "If 1,00,000 people had this, how many would be alive in one year?". "Believe in miracles," they said. And he did chance across one when they enlisted him for the clinical trial of an experimental drug, bevacizumab. "Nothing was proven at that point, but I had nothing to lose," he says. Twenty-nine months later, with clear mms and no recurrences, he is facing life with a bit more confidence.

Patients like him demonstrate how quickly cancer treatment is changing. That change is clear to veterans who have been in and out of cancer wards for long. As a student at

AIMS in the '70s, Rath remembers beaming X-rays mostly on paediatric tumours "to make those behave better". But those were high-energy electromagnetic waves that passed through the body, depositing energy all along, not just on the tumour. "From 2004-05, the next wave of radiation technology—with focused beams, targeting a well-defined tumour—started coming in," says Rath. "They help doctors to locate the cancer, know how far it has spread and beat it down in a versatile, accurate and precise manner." In his unit at AIMS, Rath now has all the latest devices crammed under its roof: PET-CT scans to see changes in the activity of cells, state-of-the-art Intensity Modulated Radiation Therapy (IMRT) machines to focus radiations on cancer-affected areas only, Image Guided Radio Therapy (IGRT) devices to deliver correct doses even if the tumour cells move during the treatment, stereotactic, proton and particle radio systems to shoot beams into tumours at the speed of light.

"A patient's quality of life has improved dramatically," points out Dr Harit Chaturvedi, chief consultant and director, surgical oncology, Max Healthcare, Delhi. That's because, all new developments—from high-end scanners and X-rays, immunohistochemistry (to analyse and identify cancer cell types), targeted chemotherapy, organ preservation to minimally invasive surgery—are focussed on power and precision to reduce patient trauma. "The problem with radiation or chemotherapy is, while they prevent cancers from spreading, they also damage healthy cells, resulting in serious side effects," he says. Because cancer therapeutics are commonly delivered through the blood, only a small fraction reaches the target tumour. To obtain maximum benefit, a high dose of drug has to be administered. "A lot of effort is on to make the drugs more tumour-specific," says Chaturvedi.

The god is in biology. Oncology is no longer a one-size-fits-all field. "We are now entering an era of personalised cancer medicines, in which



DR RAKESH JALALI (CENTRE) AT A RADIATION SESSION ON THE STATE-OF-THE-ART TRILOGY MACHINE AT TATA MEMORIAL HOSPITAL IN MUMBAI



ANURAG BASU 35

LEUKAEMIA SURVIVOR

He was diagnosed with acute leukaemia in 2004 and was told by doctors that he had little chance of surviving. "My wife was eight months pregnant and I was determined to live and see my child," he says. For the next one-and-a-half years, Basu battled cancer through chemotherapy and refused to let it take over his life. He began shooting for *Gangster* during his final dose of chemotherapy and penned *Life in a... Metro*. Looking back, he says the experience changed him a lot. "You learn the value of life when you see death closely. My first priority used to be work, but now it is my family," he says.

treatment is tailored to the unique genetics of the individual or the tumour," says Dr Rakesh Jalali, neuro-oncologist with TMC. Even in the early-90s, when he was an MD at PGIMER in Chandigarh, cancer biology remained shrouded in mystery. "A breast cancer was just a breast cancer; a glioma just that. There was a blanket therapy." Now researchers know that each cancer is different and the effort is to treat a tumour according to a whole set of personalised parameters—age of the patient, the stage, grade and type of the tumour, the pathology and possibility of recurrence and so on.

Clinical trials are the engine driving cancer research today. They make it possible to demystify cancer biology. "Globally, the investment in cancer research is enormous," says Venkatesh V. Reddy, senior pharma analyst with PharmARC Analytic Solutions in Bangalore. In the US, 800